Dual-Channel Pete Millett Interface Replica

Universal High-Voltage Active Differential Probe & Audio-Band Conditioning AFE

Target Hosts:	M-Audio Solo, MOTU M4, SDS804X HD	**Document ID:**	EDD-PMSI-UNIV-002
Topology:	Dual-Channel Galvanically Isolated AFE	**Revision:**	2.0 (Design Freeze)
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System Parameter	Hardware Implementation Target
Architecture Format	Dual-Channel (Stereo Ch 1 / Ch 2), fully mirrored analog signal paths
Active Input Buffers	Dual OPA1656 CMOS Op-Amps per channel (10 pA bias, 2.9 nV/$\sqrt{\text{Hz}}$, 53 MHz)
Overvoltage Current Limit	Back-to-back LND150 depletion MOSFETs ($\leq$ 3.0 mA fault current, 500 V rating)
Difference Amplifier Stage	Texas Instruments INA134 (Pin 1 tied to AGND, > 90 dB guaranteed CMRR)
DC Offset Trimming	Completely eliminated; laser-trimmed INA134 preserves maximum CMRR
Selectable Output Clamping	Dual-range off-signal Zener bias bridge (±1.4 V Solo mode / ±5.4 V M4 & Scope mode)
Analog Output Stages	Dual output per channel: Balanced 1/4" TRS (DRV135) + Unbalanced BNC (200 Ω)
Input Ground Switching	Permanently eliminated; full floating differential input prevents Earth shorts
AC Coupling Threshold	0.47 μF 400 V Metallized PP Film ($f_{-3\text{dB}} \approx 3.38$ Hz, multi-pitch footprint)
DC Galvanic Isolation	XP Power JCA0305D03 (3 W, 5 V in, ±15 V out, 1,500 VDC isolation, Faraday shield)
Power Input & Safety	USB-C (6-pin power) with 1812 SMD PTC fuse + SMCJ6.0A TVS crowbar clamp
Bench Host Versatility	M-Audio Solo (Audio FFT), MOTU M4 (Pro Audio), Siglent SDS804X HD (Scope)

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1 Executive Design Summary & Tri-Host Architecture

This document establishes the electrical, mechanical, and PCB layout specifications for an optimized dual-channel replica of the classic Pete Millett Soundcard Interface. While originally conceived as a single-channel, through-hole audio interface accessory for consumer PC soundcards, this Revision 2.0 architecture transforms the design into a **high-precision, dual-channel, galvanically isolated active differential probe and signal conditioning front-end**.

The system is engineered to serve three distinct instrumentation hosts interchangeably without hardware reconfiguration:

1. **M-Audio Solo (Consumer USB Audio Interface):** Provides basic single-ended and balanced audio-band spectrum analysis, serving as an isolated, sacrificial signal source and preliminary digitizer.
2. **MOTU M4 (Professional 24-bit/192 kHz Interface):** Connects via dedicated rear-panel balanced Line Inputs 3 & 4 (+18 dBu headroom, fixed analog gain) to perform high-resolution automated harmonic distortion (THD+N down to −106 dB), intermodulation, and phase tracking using software suites such as ARTA (STEPS, LIMP), REW, and Multitone.
3. **Siglent SDS804X HD (12-bit High-Resolution Bench Oscilloscope):** Functions as a **1, 500 VDC isolated, high-voltage active differential probe**. In High-Resolution (ERES) mode, the oscilloscope achieves 14 to 16 effective bits (ENOB) with continuous bandwidth from DC up to > 500 kHz, making it ideal for inspecting ultrasonic oscillation, power-supply rail ripple, slew-rate transients, and Class-D amplifier switching noise without ground-loop hazards.

1.1 Key Design Modifications from Baseline

- **Dual-Channel Mirrored Topology:** Complete duplication of attenuators, LND150 limiters, OPA1656 buffers, INA134 difference stages, clamping bridges, and DRV135 drivers for independent stereo operation.
- **Replacement of Noisy 50 kΩ Resistors:** Substituted Pete's original 49.9 kΩ resistors (R_{16}, R_2) with back-to-back Microchip LND150 depletion MOSFETs, lowering the front-end thermal noise floor by ≈ 14 dB.
- **Elimination of Input Ground Switch (SW5) & Monitor Switch (SW3):** Hardwired as a permanently floating differential front-end to eliminate the risk of accidental short-circuits to Mains Earth Ground when paired with a bench oscilloscope.
- **Elimination of DC Zero-Adjust Potentiometer (R_{25}):** Tied INA134 Pin 1 directly to local isolated analog ground (AGND). This prevents trimming resistance from unbalancing the internal laser-trimmed resistors, preserving the full > 90 dB Common-Mode Rejection Ratio (CMRR).
- **Dual-Level Off-Signal Zener Clamping Bridge:** Redesigned the output protection network with a 2-position switch/jumper selecting between $\pm 1.4\,V_{pk}$ (Solo protection) and $\pm 5.4\,V_{pk}$ (M4 line level / Scope dynamic range expansion) without placing capacitive Zener diodes on the audio signal path.
- **Universal Output Connectivity:** Simultaneous balanced 1/4″ TRS jacks (via DRV135) and unbalanced 50 Ω BNC jacks (via low-impedance tap from the protected node).

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2 System Architecture & Host Comparison

To understand how the interface integrates with various bench instruments, Table 1 outlines the operating modes and trade-offs across the supported hosts.

Metric	M-Audio Solo	MOTU M4	Siglent SDS804X HD (Hi-Res)
Primary Role	Portable / Sacrificial Audio In	Reference Audio Distortion Rig	High-Voltage Floating Active Probe
Usable Bandwidth	10 Hz – 21.5 kHz	10 Hz – 90 kHz	**DC – 500 kHz+** (up to 70 MHz raw)
Effective Bits	16-bit (ENOB ≈ 14.5)	24-bit (ENOB ≈ 19.5)	12-bit native; **14.5 – 16** bits in ERES
THD+N Noise Floor	−90 dB (0.003%)	**−106** dB (0.0005%)	−86 dB to −90 dB (0.003%)
Interface Input	Front 1/4″ Line (Bal/Unbal)	Rear 1/4″ TRS Line 3/4 (Bal)	Front BNC (1 MΩ ∥ 15 pF)
Ground Reference	Floating (USB laptop isolated)	Floating (USB bus powered)	**Mains Earth Grounded** (Chassis BNC)
Interface Output	Unbal BNC or Bal TRS	Balanced 1/4″ TRS (DRV135)	Unbalanced BNC Tap (R_{28})
Clamp Mode	Low (±1.4 V_{pk})	High (±5.4 V_{pk})	High (±5.4 V_{pk})

Table 1: Hardware capability and routing matrix across target instrumentation hosts.

2.1 Why the Oscilloscope Cannot Fully Replace the Soundcard

While the SDS804X HD in High-Res mode achieves an effective resolution of ≈ 15 bits with wide bandwidth, it cannot replace a dedicated audio interface for precision distortion testing:

- **Delta-Sigma Noise Shaping:** Audio converters (such as the AKM/ESS chips in the M4) oversample and push quantization noise outside the audio band, yielding an ultra-quiet floor down to −115 dBV to −130 dBV.
- **Wideband Johnson Noise:** The oscilloscope front-end must maintain megahertz flat bandwidth. Even when band-limited, wideband thermal noise keeps the scope's noise floor near −88 dB. Op-amp or amplifier harmonics sitting at 0.0005% (−106 dB) disappear entirely beneath the scope's quantization floor.
- **Software Sweep Automation:** ARTA (STEPS) and REW automate multi-point stepped-sine distortion vs. frequency and power sweeps, automatically tracking harmonics HD2 through HD10. Oscilloscopes require manual data export and post-processing for similar audio metrics.

2.2 Why the Oscilloscope Requires This Active Front-End

Conversely, connecting an oscilloscope directly to an audio circuit via a standard 10X passive probe presents severe limitations:

a. The Mains Earth Short Hazard: The outer shield of the scope's BNC jack is hardwired to Mains Earth. Clipping a probe ground lead to a floating tube cathode resistor, a bridge-tied load (BTL) amplifier output, or an inverter rail creates a dead short to earth ground, instantly destroying circuit traces or the scope probe. The interface's 1,500 VDC galvanic isolation completely eliminates this hazard.

b. High DC Bias Dynamic Range Loss: Probing a 300 VDC vacuum tube plate carrying a 100 mVAC signal forces a standard scope to a 50 V/div or 100 V/div scale. The 100 mV audio ripple occupies a fraction of a single ADC quantization level. The interface's 400 V polypropylene film capacitors block the DC component completely, allowing the scope to operate on its most sensitive 20 mV/div or 50 mV/div scale.

c. Capacitive Loading on High-Z Nodes: A 10X passive probe has ≈ 15 pF of tip capacitance, which at 20 kHz represents an impedance of only 530 kΩ. On a high-impedance tube plate (100 kΩ), this causes high-frequency roll-off and phase shift. The interface's OPA1656 CMOS inputs buffer the node with negligible capacitance and 10^{13} Ω input resistance.

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3 DC Power Supply & Primary Crowbar Architecture

The power delivery network converts an unregulated +5 VDC supply into low-noise, fully isolated ±15 VDC analog split rails. Because the interface is powered via standard external USB power adapters or host PC ports, robust primary-side protection is mandatory to safeguard against overvoltage transients and polarity reversals.

3.1 Power Input Connector (USB-C 6-Pin)

Legacy full-sized USB-B and coaxial barrel jacks are deprecated in favor of a modern 6-pin USB-C receptacle (Power Only).

- **Connector Pinout:** Provides two V_{BUS} pins, two Ground pins, and two Configuration Channel (CC1/CC2) pins.
- **CC Logic Configuration:** To ensure universal enumeration with standard USB Type-C to Type-C smart power supplies (E-marked cables), CC1 and CC2 must each be terminated to ground through an independent 5.1 kΩ ± 1% pulldown resistor.
- **Solderability:** The 6-pin layout provides wide (> 0.8 mm) pad spacing, permitting rapid, high-yield hand soldering compared to 24-pin high-speed data receptacles.

3.2 Active Crowbar Protection Subsystem

To prevent catastrophic failure should the user connect an off-spec power source (e.g., 9V, 12V, or 19V laptop chargers), an explicit two-stage crowbar protection circuit is positioned upstream of all active regulators.

a. Series Overcurrent Limiter (F1): Upgraded from the legacy radial leaded RUSBF110 to an 1812 surface-mount resettable PTC fuse ($I_{\text{hold}} = 1.1$ A, $I_{\text{trip}} = 2.2$ A, $V_{\text{max}} \geq 16$ V). The 1812 package provides low internal resistance ($< 0.1\,\Omega$) and planar thermal dissipation.

b. Shunt Clamping Diode (D9): Replaces the axial 1.5KE9.1A with an SMD DO-214AB (SMC) Transient Voltage Suppressor (TVS), specifically the **SMCJ6.0A** (or SMCJ6.5A).

- Reverse Standoff Voltage (V_{RWM}): 6.0 V.
- Breakdown Voltage (V_{BR}): 6.67 V to 7.37 V.
- Peak Pulse Power Dissipation: 1500 W.

Engineering Justification: The original 9.1V breakdown diode allows dangerous voltage excursions before conduction, stressing the DC-DC input stage. A 6.0V standoff diode clamps immediately above nominal 5V, driving F1 into high-impedance latch-off within milliseconds.

3.3 Isolated DC-DC Converter (XP Power JCA0305D03)

- **Key Characteristics:** 3 W continuous rating, nominal +5 VDC input (4.5 V to 9.0 V operating window), fully regulated ±15 VDC dual output at ±100 mA.
- **Galvanic Isolation:** 1,500 VDC baseline isolation breaks ground continuity between the computer sound card and floating device-under-test (DUT) circuits.
- **EMI/RFI Containment:** Housed in an integrated nickel-coated copper metal enclosure. The conductive shell acts as an intrinsic Faraday cage, trapping high-frequency magnetic and electric fields.
- **Switching Profile:** Fixed switching frequency at 300 kHz. Because 300 kHz sits over an order of magnitude above the audible audio spectrum (20 Hz–20 kHz) and its sampling limits (192 kHz A/D Nyquist at 96 kHz), switching intermodulation artifacts are readily rejected.

3.4 Bulk Storage and High-Frequency Filtering Network

The legacy design specified 35V-rated radial electrolytic capacitors indiscriminately across both the 5V and 15V domains, consuming excessive board area. The revised component sizing is structured as follows:

- **Input Bulk Decoupling (C22):** Downrated from 35V to 10V or 16V (330 μF, low-ESR radial or aluminum SMD). Dimensions reduce from a diameter of 10 mm to 6.3 mm.
- **High-Frequency Input Snubber:** An 0805 0.1 μF X7R ceramic capacitor is routed directly across pins 22/23 ($V_{\text{IN+}}$) and pins 2/3 ($V_{\text{IN-}}$) of the JCA module. Electrolytic bulk capacitors exhibit significant inductive reactance at 300 kHz; the local ceramic shunt captures inductive current spikes before they radiate across the input plane.
- **Output Rail Filtering (C7, C8, C13, C18):** Downrated from 35V to 25V (330 μF low-ESR). Sized for reliable headroom above the continuous ±15 VDC rail potential while minimizing footprint height.

Circuit Block	Qty	Current (±15 V)	Operating Condition / Notes
OPA1656IDR (Dual Op-Amp)	2 ICs	14.0 mA total	High-Z input buffers (Ch 1 & Ch 2)
INA134UA (Diff Amp)	2 ICs	4.8 mA total	Difference amplifiers (Ch 1 & Ch 2)
DRV135UA (Line Driver)	2 ICs	10.4 mA total	Quiescent line driving current
Zener Bias Networks (R_{29}, R_{30})	4 legs	20.6 mA total	Bias current in High-Clamp mode
Total Quiescent Draw	—	**49.8** mA	49.8% **of** 100 **mA converter rating**

Table 2: Power consumption budget across dual-channel active components.

3.5 Power Budget for Dual-Channel Architecture

Table 2 verifies that current demands for both channels remain well within continuous operating ratings of the single JCA0305D03 converter.

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4 Analog Front-End (AFE) Signal Conditioning

The input stage accepts single-ended or balanced voltages ranging from line-level audio ($< 1\,V_{RMS}$) up to high-voltage tube amplifier plate swings ($> 100\,V_{RMS}$).

4.1 High-Voltage DC Blocking Capacitors (C2, C3, C20, C21)

DC blocking capacitors isolate the sensitive solid-state buffer from the high DC anode potentials encountered in vacuum tube circuits.

- **Capacitance Optimization:** Pete Millett originally specified 1.0 μF 400V polyester dielectric blocks to extend sub-audible measurement performance down to 1.5 Hz. For standard audio analysis (20 Hz–20 kHz), the input coupling values are downsized to 0.47 μF (400VDC working rating).
- **Cut-off Calculation:** Across the 100 kΩ system input impedance, the high-pass -3 dB corner frequency is given by:
$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 10^5 \times (0.47 \times 10^{-6})} \approx 3.38\,\text{Hz} \tag{1}$$
At 20 Hz, the magnitude attenuation is below 0.06 dB, and phase shift is negligible.
- **Dielectric Selection:** Metallized Polypropylene (PP) Film (e.g., WIMA MKP4 or Panasonic ECQ-E). High-voltage Class II SMD ceramic capacitors (X7R, X5R) are strictly prohibited due to severe Voltage Coefficient of Capacitance (VCC) non-linearities, which inject harmonic distortion under high-voltage AC swings.
- **CAD Footprint Provisioning:** The PCB layout incorporates multi-pitch radial through-hole pads with concentric drill points spaced at 5.0 mm, 7.5 mm, 10.0 mm, and 15.0 mm, ensuring compatibility with various film capacitor form factors.

4.2 Active Protection Clamping & Low-Noise Current Limiting

When an overvoltage transient enters the front-end, high currents must be prevented from damaging the op-amp input stages.

4.2.1 Thermal Noise Analysis & LND150 Substitution

In Pete Millett's original design, 49.9 kΩ resistors (R_2, R_{16}) were placed in series with the input buffer to limit fault current into the protection diodes during overvoltage events.

The Johnson-Nyquist thermal noise generated by a resistor is:

$$v_n = \sqrt{4k_B T R \Delta f} \tag{2}$$

At room temperature ($T = 300\,\text{K}$), the noise spectral density of the legacy 49.9 kΩ resistor is:

$$e_{n(\text{passive})} = \sqrt{4 \times (1.38 \times 10^{-23}) \times 300 \times 49{,}900} \approx 28.7\,\text{nV}/\sqrt{\text{Hz}} \tag{3}$$

Because the modern OPA1656 operational amplifier exhibits an ultra-low input noise density of only $2.9\,\text{nV}/\sqrt{\text{Hz}}$, Pete's 50 kΩ resistors degraded the front-end noise floor by nearly an order of magnitude.

The LND150 Solution: A back-to-back pair of Microchip LND150 depletion-mode N-channel MOSFETs (500 V breakdown rating) replaces each 50 kΩ resistor:

- **Connection Topology:** The two MOSFETs are wired Source-to-Source. Both Gates are tied directly to the common Source node ($V_{GS} = 0\,\text{V}$).
- **Normal Audio Signal Conduction:** Under normal operating swings ($|V| < \pm 15\,\text{V}$), both transistors remain fully on with zero gate-source bias. The combined channel resistance ($R_{\text{DS(on)}}$) is only ≈ 1.5 kΩ to 2.0 kΩ total.
- **Thermal Noise Reduction:** The noise density drops from $28.7\,\text{nV}/\sqrt{\text{Hz}}$ to:

  $$e_{n(\text{active})} \approx \sqrt{4 \times (1.38 \times 10^{-23}) \times 300 \times 2{,}000} \approx 5.75\,\text{nV}/\sqrt{\text{Hz}} \tag{4}$$

  This achieves a 14 **dB improvement in input SNR**.
- **Overvoltage Pinch-Off Action:** When a fault voltage exceeding ±15 V strikes the input, the fault current flows through the forward-biased body diode of the upstream FET into the channel of the downstream FET. The voltage drop across the channel reverse-biases the Gate-Source junction ($V_{\text{GS}} \leq V_{\text{GS(off)}} \approx -1.8\,\text{V}$), pitching off the channel and clamping the maximum current to a safe 1.5 mA to 3.0 mA even under continuous 500 VDC fault conditions.
- **Input Rail Clamps:** Silicon high-speed switching diodes (**1N4148W**, SOD-123) shunt the limited 2 mA current into the ±15 V rails. The 1N4148W maintains a junction capacitance under 2 pF, preventing capacitive high-frequency roll-off.

4.3 Why the Input Attenuator Ladder Must Be Retained

It is tempting to consider bypassing the input resistor divider ($R_4 - R_7$) when using an oscilloscope host, since scopes have their own internal attenuators. **This is fundamentally prohibited.**

- The active OPA1656 input buffers operate on ±15 V power rails, giving them a maximum linear input voltage swing of $\approx \pm 13.5\,V_{peak}$ ($9.5\,V_{RMS}$).
- Probing a vacuum tube amplifier plate (150 V to 350 V) without attenuation would drive the OPA1656 far past its rails, forcing the 1N4148W clamping diodes to conduct continuously. The waveform delivered to the scope would be clipped into a distorted square wave at ±15.6 V.
- The precision stepped attenuator (200 mV, 2 V, 20 V, 200 V full-scale ranges) is what enables the interface to scale high-voltage potentials down to the linear operating region of the op-amps.

4.4 Rotary Range Switch Implementation

- **Switch Topology Simplification:** With the elimination of the dual-rail analog metering circuit, the original multi-wafer 2-pole switch is simplified to a single-pole four-throw (1P4T) configuration per channel.
- **Component Selection:** Alpha SR2611F 1-Pole, 12-Position rotary switch (SW4, SW4b). The included adjustable metal mechanical stop ring is locked at hole position 4, providing an exact, positive-stop 4-position action.
- **Wiring Harness:** The switches are wired using 5 active conductors per channel (1 common input lead and 4 range taps). These conductors are bundled and tightly twisted to maintain high common-mode magnetic rejection.

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5 Analog Buffer, Instrumentation Amplifier & CMRR Optimization

5.1 Operational Amplifier Selection: Texas Instruments OPA1656IDR

The dual operational amplifiers (U4, U5) serve as the high-impedance buffer stage. Precision measurements demand that the op-amp introduce negligible loading error and zero DC drift across the attenuator.

DC Offset Analysis: Across the worst-case 100 kΩ input resistance, a traditional bipolar op-amp (NE5532, $I_B \approx 72$ nA) develops an uncalibrated DC offset:

$$V_{\text{offset}} = I_B \times R_{\text{in}} = 72 \times 10^{-9}\,\text{A} \times 100 \times 10^{3}\,\Omega = 7.2\,\text{mV} \tag{5}$$

In contrast, the OPA1656 generates:

$$V_{\text{offset}} = 10 \times 10^{-12}\,\text{A} \times 100 \times 10^{3}\,\Omega = 0.001\,\text{mV} \quad (1.0\,\mu\text{V}) \tag{6}$$

This negligible DC offset preserves high DC stability without requiring manual calibration.

Parameter	NE5532 (Legacy Bipolar)	OPA2134 (Legacy JFET)	OPA1656 (Selected CMOS)
Input Bias Current (I_B)	200 nA (typ), 800 nA (max)	5 pA (typ), 100 pA (max)	**10** pA (typ), **25** pA (max)
Voltage Noise Density (e_n)	5.0 nV/$\sqrt{\text{Hz}}$	8.0 nV/$\sqrt{\text{Hz}}$	**2.9** nV/$\sqrt{\text{Hz}}$
Gain-Bandwidth Product (GBW)	10 MHz	8 MHz	**53** MHz
Slew Rate (SR)	9 V/μs	20 V/μs	**24** V/μs
THD+N (1 kHz, 2 kΩ)	0.0005%	0.00008%	**0.000029**% (−131 dB)
Input Offset Voltage Drift	2 μV/°C	2 μV/°C	**0.5** μV/°C
Package / Availability	DIP-8 / SOIC-8	DIP-8 / SOIC-8	SOIC-8 (SMD Surface Mount)

Table 3: Parametric comparison demonstrating the operational advantages of the OPA1656 CMOS amplifier.

5.2 Elimination of the DC Nulling Network (R_{25})

In the original design, a 100 kΩ cermet potentiometer (R_{25}) was connected to the reference pin (Pin 1) of the INA134 difference amplifier to null DC offsets.

Engineering Justification for Elimination:

1. **Preservation of CMRR:** The Texas Instruments INA134 relies on four internal, on-chip thin-film resistors laser-trimmed to an absolute ratio match better than 0.005%. This precise matching guarantees a Common-Mode Rejection Ratio of > 90 dB across the audio band. Injecting an external DC voltage into Pin 1 introduces source impedance non-linearities and temperature coefficients that unbalance the internal bridge, degrading CMRR by 15 dB to 25 dB.
2. **CMOS Buffer Zero Drift:** The OPA1656 features an input bias current of only 10 pA, reducing offset drift to negligible microvolt levels.
3. **Hardwired Ground Implementation:** Pin 1 of U7 and U9 is hardwired directly to the solid analog ground plane (AGND). Both target audio interfaces and oscilloscopes support software AC coupling, rendering sub-millivolt DC offsets irrelevant.

5.3 Removal of Grounding Switch (SW5) & Monitor Switch (SW3)

Pete Millett's input section included 'SW5' (which shunted the negative input V_{INN} directly to local chassis ground) and 'SW3' (which looped the signal generator back into the input). Both switches have been eliminated:

- **Ground Loop Hazard with Oscilloscopes:** When connecting the interface output to the Siglent SDS804X HD, the interface's local analog ground becomes referenced to Mains Earth Ground via the scope's BNC cable shield. If 'SW5' were closed while probing a circuit with a non-zero reference potential (such as a tube amplifier cathode bias network or a Class-D bridge-tied output), it would create a **direct short-circuit to Mains Earth**, vaporizing PCB traces or damaging the device under test.
- **Permanent Floating Differential Operation:** By completely removing 'SW5', V_{INN} is routed through DC blocking capacitor C_2 into the non-inverting input of buffer U4B. The front-end remains permanently floating and differential, providing complete galvanic safety regardless of switch settings.

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6 High-Frequency Stability, Layout & Oscillation Prevention

The Texas Instruments OPA1656 is a wide-bandwidth operational amplifier ($f_{\text{unity}} = 53\,\text{MHz}$). At these frequencies, parasitic board capacitances and load cable capacitances can erode amplifier phase margin, resulting in high-frequency ringing or sustained oscillation.

6.1 Active Layout Stability Rules

a. **Direct Rail Decoupling:** High-frequency surface-mount ceramic bypass capacitors (0.1 μF X7R, 0805) are placed directly against Pin 8 ($V+$) and Pin 4 ($V-$). Power traces route through the capacitor solder pad prior to entering the IC lead frame. A 10 μF tantalum or low-ESR electrolytic capacitor is placed within 10 mm upstream on each rail.

b. **Feedback Phase Compensation ($C_{\mathbf{fb}}$):** An 0805 capacitor footprint is routed in parallel with the main feedback resistor (R_{fb}). This footprint is populated with a 10 pF to 22 pF C0G/NP0 dielectric ceramic capacitor. This small feedback capacitance introduces a high-frequency zero in the loop transmission, providing phase lead to stabilize the internal pole against parasitic input capacitance.

c. **Series Output Isolation (R_{28}):** Driving capacitive coaxial cables directly from low-impedance op-amp outputs degrades phase margin according to $f_{\text{pole}} = \frac{1}{2\pi R_o C_{\text{load}}}$. A dedicated 200 Ω thin-film resistor (R_{28}) isolates the INA134 output from cable capacitance.

6.2 PCB Floorplan & Partitioning Strategy

- **Galvanic Isolation Barrier:** A strict 4.0 mm clearance moat separates the non-isolated USB-C 5 V input domain from the isolated ±15 V analog domain. No copper traces, ground planes, or component pads bridge this barrier except for the footprint of the JCA0305D03 module.
- **Mirrored Symmetry:** Channel 1 and Channel 2 analog signal paths are routed with identical trace lengths and parasitic capacitances, placed symmetrically north and south of a central analog ground bus.
- **Isolated Ground Plane (AGND):** The secondary side of the board utilizes an unbroken, contiguous copper pour on Layer 2 dedicated exclusively to isolated analog ground. This provides low-impedance return paths and minimizes channel-to-channel crosstalk to below −110 dB at 1 kHz.

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7 Dual-Level Output Clamping & Driver Subsystem

7.1 Why Zener Diodes Cannot Be Placed in the Signal Path

When designing an overvoltage protection clamp for an audio measurement system, clamping diodes cannot simply be connected across the audio line:

- **Junction Capacitance Non-Linearity:** Standard Zener diodes (e.g., 5.1 V Zeners) exhibit massive junction capacitances ($C_j \approx 100\,\text{pF}$ to 400 pF). This capacitance varies non-linearly with applied signal voltage, generating significant harmonic distortion (THD).

- **Soft Reverse Breakdown Knee:** Zener diodes begin conducting microamps of leakage current well below their nominal breakdown rating. On audio peaks above 1 V_{RMS}, this leakage current clips signal peaks roundedly, introducing odd-order harmonics into measurements.

7.2 The Off-Signal Zener Bias Bridge Topology

To achieve high-voltage clamping with zero audio-band distortion, this design keeps ultra-fast, low-capacitance switching diodes (**1N4148W**, $C_j < 2$ pF) on the audio signal path while relocating the voltage-defining Zener diodes into the **DC power rail bias legs**.

Positive Bias Rail (Node A):
+15 V $\longrightarrow$ [1 kΩ (R_{29})] $\longrightarrow$ **Node A**

Node A $\longrightarrow$ [Switch / Jumper] $\longrightarrow$
- D_{11} (1N4148) $\longrightarrow$ AGND — (Solo: $V_A \approx +0.7$ V)
- DZ_1 (4.7V Zener) $\longrightarrow$ AGND — (M4/Scope: $V_A \approx +4.7$ V)

Audio Clamping Path:
INA134 Out $\longrightarrow$ [200 Ω (R_{28})] $\longrightarrow$ **SCPROT** $\longrightarrow$ Outputs
Cathode(D_{10}) $\longleftarrow$ **Node A** | Anode(D_{10}) $\longleftarrow$ **SCPROT**

Negative Bias Rail (Node B):
−15 V $\longrightarrow$ [1 kΩ (R_{30})] $\longrightarrow$ **Node B**

Node B $\longrightarrow$ [Switch / Jumper] $\longrightarrow$
- D_{13} (1N4148) $\longrightarrow$ AGND — (Solo: $V_B \approx -0.7$ V)
- DZ_2 (4.7V Zener) $\longrightarrow$ AGND — (M4/Scope: $V_B \approx -4.7$ V)

Anode(D_{12}) $\longleftarrow$ **Node B** | Cathode(D_{12}) $\longleftarrow$ **SCPROT**

7.2.1 Circuit Operation and Clamping Thresholds

The signal node ('SCPROT') connects to the anodes and cathodes of steering diodes D10 and D12. Under normal operation, D10 and D12 are reverse-biased by Node A ($+V_{\text{bias}}$) and Node B ($-V_{\text{bias}}$). Because the diodes are fully reverse-biased, they inject zero non-linear leakage into the audio line and present less than 2 pF of total shunt capacitance.

A DPDT toggle switch (SW1) or 3-pin hardware jumper block configures the bias nodes:

1. **Position 1: Low Range (M-Audio Solo Mode):**
   - Node A is clamped by diode D11 ($V_f \approx +0.7$ V). Node B is clamped by diode D13 ($V_f \approx -0.7$ V).
   - When an overvoltage occurs, D10 conducts once $V_{\text{signal}} > V_A + V_{f(\text{D10})} \approx 0.7\,\text{V} + 0.7\,\text{V} = +\mathbf{1.4}\,\text{V}_{\text{peak}}$.
   - Negative swings clamp at $-\mathbf{1.4}\,\text{V}_{\text{peak}}$ via D12 and D13.
   - Full-scale output is limited to $\approx 1.0\,\text{V}_{\text{RMS}}$, protecting the 16-bit M-Audio Solo ADC from overdriving and digital clipping.
2. **Position 2: High Range (MOTU M4 & Oscilloscope Mode):**
   - Switch SW1 routes Node A through a 4.7 V Zener diode (DZ_1, BZX84C4V7) and Node B through DZ_2.
   - Bias voltages stabilize at $V_A = +4.7$ V and $V_B = -4.7$ V.

- The signal clamps at $V_{\text{signal}} = \pm(4.7\,\text{V} + 0.7\,\text{V}) = \pm\mathbf{5.4}\,\text{V}_{\text{peak}}$ ($\approx \mathbf{3.8}\,\text{V}_{\text{RMS}}$, or $+\mathbf{14.5}\,\text{dBu}$).
- This higher threshold takes full advantage of the MOTU M4's +18 dBu max input headroom and provides a high signal-to-noise ratio into the SDS804X HD.

7.2.2 Bias Resistor Sizing and Noise Suppression

- **Current Calculation:** On the ±15 V split rails with 4.7 V Zeners active:

$$I_{\text{bias}} = \frac{15\,\text{V} - 4.7\,\text{V}}{1{,}000\,\Omega} = 10.3\,\text{mA} \tag{7}$$

- **Power Dissipation:** Resistors R_{29} and R_{30} dissipate $P = (10.3\,\text{mA})^2 \times 1\,\text{k}\Omega = 106\,\text{mW}$. Standard 0805 or 1206 0.25 W resistors run cool with high reliability. Zener power dissipation is $4.7\,\text{V} \times 10.3\,\text{mA} = 48\,\text{mW}$, well below standard 350 mW ratings.
- **Avalanche Noise Shunting:** Zener diodes generate wideband avalanche shot noise. To prevent this noise from coupling into the bias nodes and modulating the steering diodes, a 0.1 μF X7R ceramic bypass capacitor is placed in parallel across each Zener diode (C_{z1}, C_{z2}).

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8 Universal Dual Output Stage: DRV135 & BNC Tap

Both channels feature two parallel, simultaneously active output connections:

a. Unbalanced BNC Jack: Directly tapped from the protected node ('SCPROT') through series current-limiting resistor R_{28} (200 Ω). This connector feeds directly into the SDS804X HD oscilloscope or unbalanced line-in soundcards.

b. Balanced $1/4''$ **TRS Jack:** Driven by a Texas Instruments **DRV135UA** balanced line driver (SOIC-8). The input of the DRV135 (Pin 4) taps the low-impedance 'SCPROT' node. The DRV135 provides an exact laser-trimmed differential gain of 2× (+6 dB), driving long balanced shielded cables into the MOTU M4 rear Line Inputs 3 & 4 with high common-mode ground noise rejection.

Instrument	Primary Role	Connection Mode	Clamp Range Setting
M-Audio Solo	Basic audio FFT sweeps, isolated signal source	Rear BNC to 1/4" Line In or DRV135 to Ch 2 TRS	Solo Mode (±1.4 V_{pk})
MOTU M4	Precision 24-bit/192 kHz audio analyzer	DRV135 TRS to Rear Line Inputs 3 & 4	M4 / Scope Mode (±5.4 V_{pk})
SDS804X HD	High-voltage active floating probe (DC–500 kHz+)	Rear BNC direct to Scope 1 MΩ Channel	M4 / Scope Mode (±5.4 V_{pk})

Table 4: Multi-host connection and configuration schedule.

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9 System Chassis Wiring & Mechanical Integration

To ensure high-frequency stability with the 53 MHz OPA1656 and preserve high CMRR, off-board panel controls must follow strict wiring practices:

Circuit Path	Conductor Type	Routing Specification
Main DC Power (V_{BUS}, GND)	18 AWG Multi-strand Copper	Direct route from USB-C breakout to PCB power terminal; minimizes DC IR drop.
Analog Audio Inputs & Outputs	24 AWG or 26 AWG Shielded Coax	Miniature RG-178 or high-flex shielded audio cable; drain shield tied at PCB star point.
Rotary Attenuator Controls	24 AWG Discrete Twisted Harness	Tightly twisted bundle ($\geq$ 3 twists per cm) to ensure external magnetic fields cancel.
Enclosure Chassis Ground	20 AWG Solid Copper	Direct bond to aluminum chassis star washer lug adjacent to USB-C ground plane.

Table 5: Interconnect and chassis wiring schedule.

9.1 Chassis Construction and RF Shielding

The complete assembly must be enclosed in an extruded aluminum enclosure (e.g., Hammond 1455 series). The aluminum chassis connects to signal ground at a single star point adjacent to the power input. All anodized layers must be cleared with a star washer to ensure low contact resistance between panels and chassis ground. Isolated Teflon front BNC connectors maintain the floating front-end reference.

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10 Consolidated Hardware Bill of Materials (BOM)

Designator	Component Function	Selected Part	Package / Specification
U1	Isolated DC-DC Converter	**JCA0305D03**	3 W, 5 V In, ±15 V Out, 1.5 kV Isolation
U4, U5	Dual Input Buffer (Ch 1, 2)	**OPA1656IDR**	SOIC-8, CMOS, 53 MHz, 2.9 nV/$\sqrt{\text{Hz}}$
U7, U9	Difference Amp (Ch 1, 2)	**INA134UA**	SOIC-8, Laser-Trimmed > 90 dB CMRR
U8, U10	Balanced Line Driver (Ch 1, 2)	**DRV135UA**	SOIC-8, Low-Distortion Line Driver
Q1–Q8	Overvoltage Current Limiter	**LND150N8-G**	SOT-89 Depletion MOSFET, 500 V rated
D1–D8	Input Clamping Diodes	**1N4148W**	SOD-123, Fast Switching, $C_j < 4$ pF
D_{10}–D_{17}	Output Pre-Biased Clamp	**1N4148W**	SOD-123, Reverse-biased low-capacitance clamp
DZ_1–DZ_4	Voltage Rail Zeners	**BZX84C4V7**	SOT-23, 4.7 V 350 mW Zener Diode
F1	Resettable Power Fuse	**SMD1812-110**	1812 SMT, 1.1 A Hold, 2.2 A Trip, 16 V
D9	Primary Crowbar TVS	**SMCJ6.0A**	DO-214AB, 6.0 V Standoff, 1500 W Transient

Table 6: Active Silicon, Power & Protection Bill of Materials (Dual-Channel).

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Designator	Component Function	Selected Part	Package / Specification
C2, C3, C20, C21	High-Voltage DC Blocking	**0.47 μF 400 V**	WIMA MKP4 Metallized Polypropylene Film
C22	Primary Bulk Input Decoupling	**330 μF 16 V**	Low-ESR Radial / Aluminum Electrolytic
C7, C8, C13, C18	Output Rail Bulk Filter	**330 μF 25 V**	Low-ESR Radial / Aluminum Electrolytic
C_{snub}	DC-DC Switching Snubber	**0.1 μF 50 V**	0805 X7R Ceramic, across JCA input pins
C_{z1}–C_{z4}	Zener High-Frequency Bypass	**0.1 μF 50 V**	0805 X7R Ceramic, in parallel with Zeners
R_4–R_7, R_{4b}–R_{7b}	Precision Attenuator Ladders	**0.1% Thin Film**	90.9 kΩ, 9.09 kΩ, 909 Ω, 102 Ω (0805)
R_{28}, R_{28b}	Output Isolation / Impedance	**200 Ω 1%**	0805 Thin Film, Native Unbalanced Tap
$R_{29,30}$, $R_{29b,30b}$	Clamp Pre-Bias Resistors	**1 kΩ 1% 0.25W**	1206 / 0805 Thin Film
SW4, SW4b	Attenuator Range Switches	**Alpha SR2611F**	1-Pole 12-Position Rotary (Stopped at Pos 4)
SW1A, SW1B	Dual Clamp Range Selector	**DPDT Sub-Mini**	Panel Toggle / On-Board 3-Pin Jumpers
J1, J2	Signal Input Connectors	**Amphenol BNC**	Chassis Mount Isolated Floating BNC
J3, J4	Unbalanced Scope Outputs	**Isolated BNC**	Chassis Mount BNC, Tapped from R_{28}
J5, J6	Balanced Audio Outputs	**1/4" TRS Jack**	Neutrik / Rean NYS216 Isolated Jack
J7	DC Power Input	**USB-C 6-Pin**	Surface Mount Power-Only with 5.1 kΩ pulldowns

Table 7: Passive Components & Electro-Mechanical Hardware Bill of Materials.

11 Bench Commissioning & Verification Workflow

Once fabricated, the dual-channel interface should be validated following this step-by-step procedure:

1. **Cold Resistance & Isolation Verification:** With power disconnected, verify $> 100\,\text{M}\Omega$ insulation resistance between the USB-C shield and the analog BNC shells using a high-voltage insulation tester or high-range DMM. Ensure Pin 1 of both INA134 ICs measures $< 0.1\,\Omega$ to isolated AGND.

2. **Power Rail Check:** Apply +5 VDC via USB-C. Verify that the JCA0305D03 output rails measure precisely between ±14.85 V and ±15.15 V with under 5 $\text{mV}_{\text{p-p}}$ switching ripple on the oscilloscope.

3. **Output Clamping Calibration:**

   - Set SW1 to Low (Solo Mode). Inject a 1 kHz sine wave into the front BNC and increase generator level. Observe on the SDS804X HD that the waveform begins symmetrical hard clipping at precisely $\pm 1.40\,\text{V}_{\text{peak}} \pm 0.05$ V.
   - Flip SW1 to High (M4/Scope Mode). Verify that the clipping threshold shifts upward to $\pm 5.40\,\text{V}_{\text{peak}} \pm 0.15$ V without distortion on signals below $3.5\,\text{V}_{\text{RMS}}$.

4. **Host Connection Protocols:**

   - *M-Audio Solo:* Connect rear BNC out to Solo Input 2 set to 'Line', with clamp switch set to **Low**.
   - *MOTU M4:* Connect rear $1/4''$ TRS jacks to rear Line Inputs 3 & 4 via TRS-to-TRS balanced cables, with clamp switch set to **High**.
   - *Siglent SDS804X HD:* Connect rear BNC jacks to Scope Ch 1 and Ch 2 via 50 Ω coaxial cables, with clamp switch set to **High** and scope vertical channel set to 1 MΩ input impedance.

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12 Summary & Next Steps

This updated specification establishes a low-noise, dual-channel active differential probe and conditioning front-end serving the M-Audio Solo, MOTU M4, and SDS804X HD.

The immediate follow-on engineering tasks are:

1. Generate the complete schematic netlist in KiCad including all unified power, protection, attenuator, and buffer sheets.
2. Finalize the mechanical drilling template for the Hammond extruded enclosure front/rear panels.
3. Execute PCB routing following the strict 2-layer star-grounding and high-frequency decoupling layout guidelines outlined in Section 5.